

PARTS LIST

Semiconductors:

- IC1 - AT89C51 microcontroller
- IC2 - 7805, 5V regulator
- T1 - BC347 npn transistor
- D1-D4 - 1N4007 rectifier diode
- LED1-LED5 - 5mm LED
- LCD - 2-line/16-character LCD

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3-R7 - 470-ohm
- R2 - 10-kilo-ohm
- R8 - 100-ohm
- VR1 - 10-kilo-ohm
- RNW1, RNW2 - 10-kilo-ohm

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2 - 0.1 μ F ceramic disk
- C3 - 10 μ F, 16V electrolytic
- C4, C5 - 33pF ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 9V, 500mA secondary transformer
- S1-S9 - Push-to-on tactile switch
- X_{cr} - 12MHz crystal
- PZ1 - Piezobuzzer

letters of the message can be encoded. Every time you select a letter, it is confirmed by pressing S7 (OK) key.

The message can consist of letters, numbers and special characters. For inputting numbers, press switch S5 (NU.). Select the desired number by using key S2 or S3. Confirm the selection by pressing S7 (OK) key.

For inputting special characters (only nine special characters have been programmed in this encoder), press key S6 (SP.- CH.). Select the de-

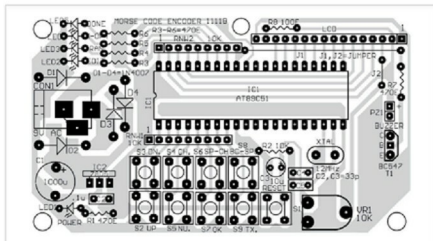


Fig. 5: Component layout for the PCB

sired special character by using either key S2 or S3. Confirm the selected special character by pressing key S7 (OK). Every time key S7 is pressed, the cursor moves forward by one digit. To move the cursor backward by one digit, press key S8 (BC.-SPC.). After the entire message is encoded, it can be heard as an audible tone from piezobuzzer PZ1 by pressing transmit key S9 (TX).

Edit mode is indicated by LED2, while transmit mode is indicated by LED3. After the desired message is transmitted, LED5 glows momentarily to indicate that transmission of the desired message is complete. LED4 glows for 500 milliseconds

when a 'dot' is transmitted and for 1500 milliseconds when a 'dash' is transmitted.

Software

The software for the microcontroller-based Morse code encoder is written in 'C' language and compiled using the Keil μ Vision4 compiler. The program is well-commented and easy-to-understand. Burn the generated hex code into the microcontroller by using a suitable programmer.

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MORSE.C

```
#include<reg51.h>
char data *p=0x30;
char data *q=0x30;
char data *r=0x40;
int a,b,c,d,e,f,g,h,i,j;
sbit up=P1^0;
// keyboard initialization////
sbit dn=P1^1;
sbit ch=P1^2;
sbit no=P1^3;
sbit up=P1^4;
sbit ok=P1^5;
sbit cl=P1^6;
sbit ts=P1^7;
sbit rs=P0^0;
// LCD initialization////
sbit rw=P0^1;
sbit en=P0^2;
sbit edl=P0^3;
// LED array initialization////
sbit tkl=P0^4;
sbit ml=P0^6;
sbit dl=P0^5;
sbit hr=P0^7;
void lsd_inti();
void dat();
void dy();
void dlt();
```

```
void dah();
void dot_delay();
void main()
{
    edl=0;
    tkl=0;
    ml=0;
    dl=0;
    hr=0;
    lsd_inti();
    P2=0x01;
    cmd();
    P1=0x0ff;
    a=0x01;
    d=0x30;
    e=0x20;
    b=0x80;
    P2=02;
    while(1)
    {
        dl=0;
        edl=1;
        if(ch==0) // check whether key (S4) is
        pressed for entry of alphabets//
        {
            a=c;
            P2=a;
            dat();
        }
    }
}
```

```

    }
    if(no==0) // check whether key (S5) is
    pressed for entry of numbers//
    {
        a=d;
        P2=a;
        dat();
    }
    if(sp==0) //check whether key (S6) is
    pressed for entry of special characters//
    {
        a=e;
        P2=a;
        dat();
    }
    if(up==0) // check whether key
    (S2) is pressed for INCREMENTING KEY VAL-
    UES //
    {
        P2=b;
        cmd();
        if(a==0x3b)
        {
            a=c;
        }
    }
}
```